

Tren Computer Science - AI, IoT, Blockchain

- CAK2MDB2
- WAWASAN GLOBAL TIK
- MATERI SUPLEMEN I
- DOSEN:
 - Aji Gautama Putrada

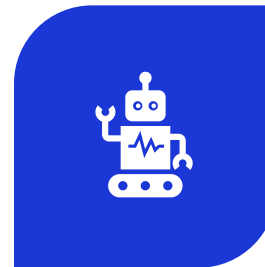
Revolusi Teknologi di Computer Science



DARI KOMPUTER PERTAMA (ENIAC) HINGGA KOMPUTASI KUANTUM DAN AI MODERN.



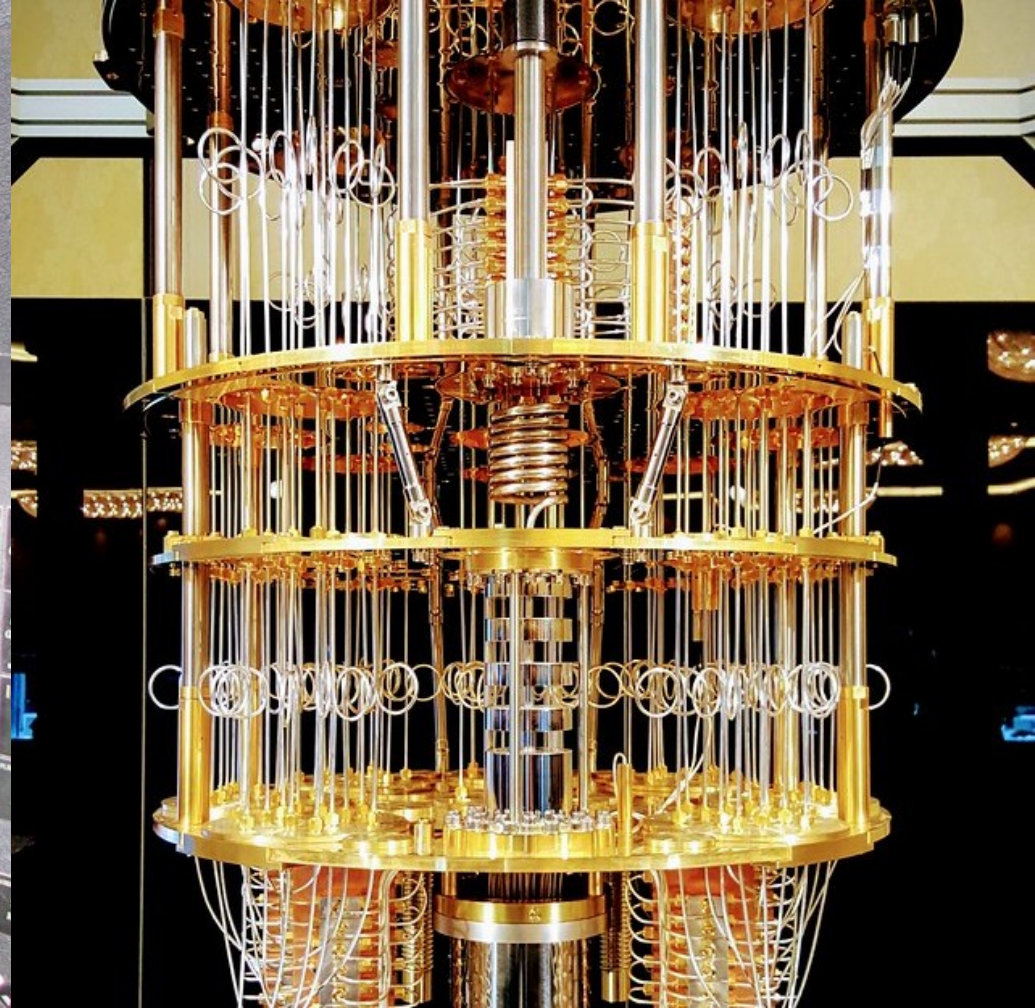
DAMPAK *MOORE'S LAW* DALAM PENINGKATAN DAYA KOMPUTASI.



AI DALAM OTOMASI INDUSTRI, IOT DALAM SMART CITIES, DAN BLOCKCHAIN DALAM KEAMANAN DATA.

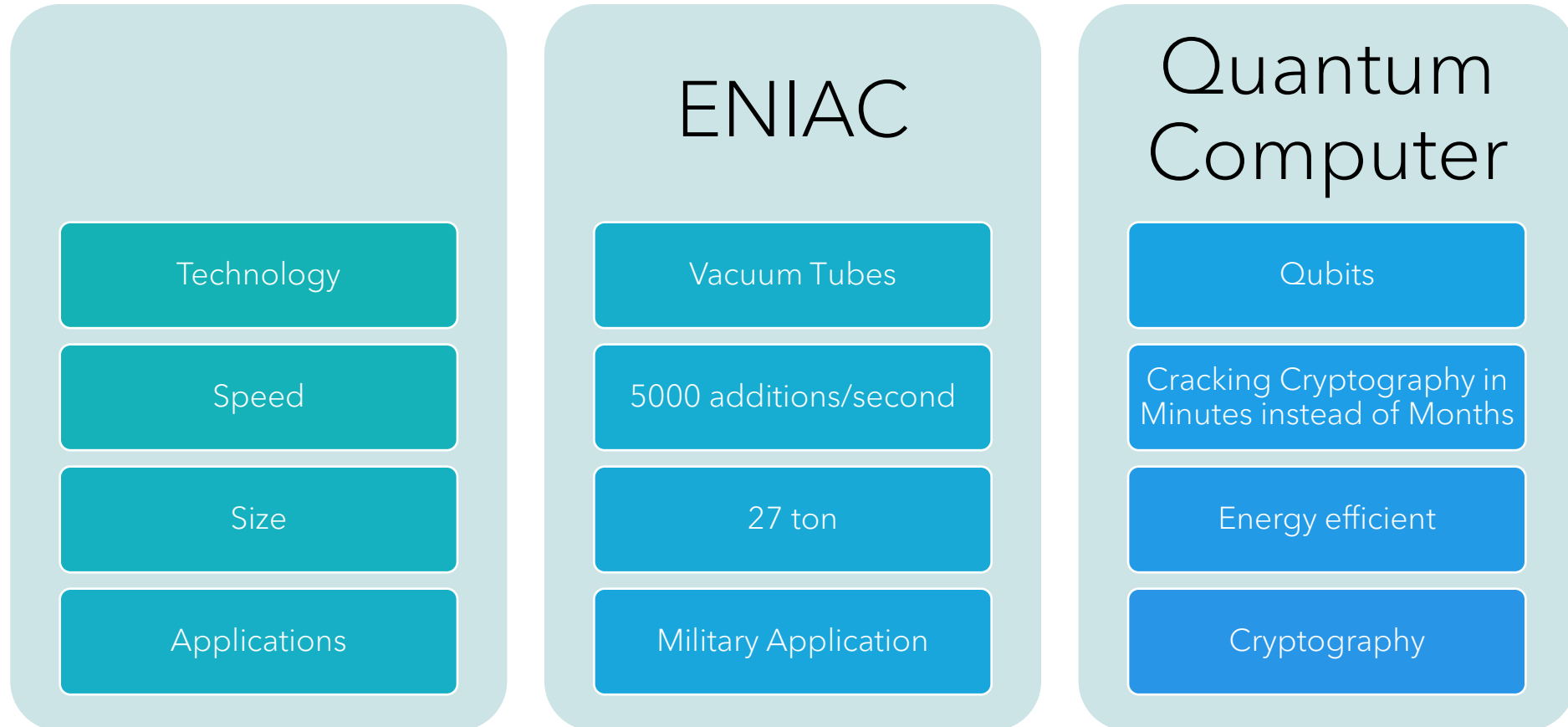


KONVERGENSI TEKNOLOGI (AI, IOT, BLOCKCHAIN, CLOUD)



Eniac vs Quantum Computer

ENIAC vs Quantum Computer



Moore's Law

The Law

The number of transistors on a chip doubles approximately every two years, (exponential growth)

Golden Era of Scaling (1970s-2000s):

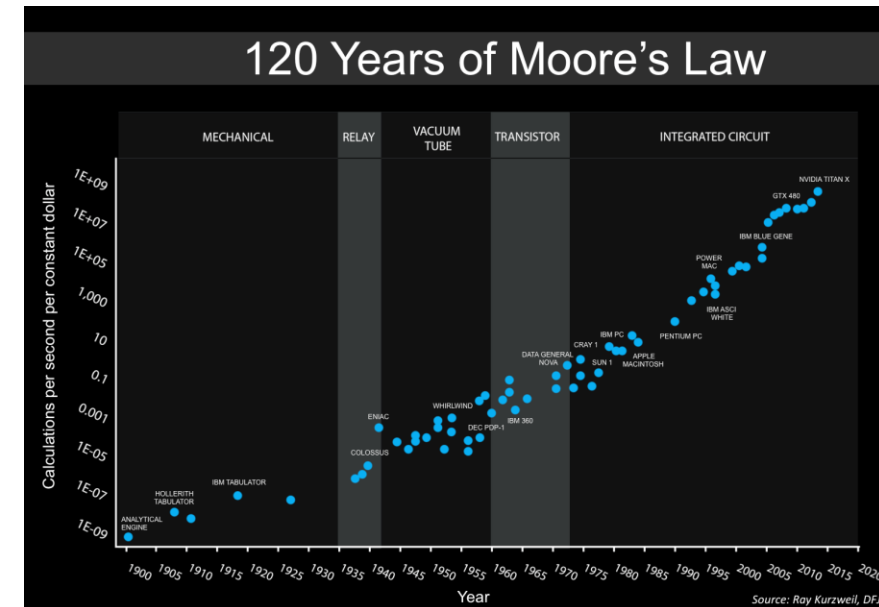
Advancements in silicon fabrication

Slowdown & Challenges (2010s-Present)

The focus shifted to parallel computing, GPUs, and TPUs.

Beyond Moore's Law - The Future

3D chip stacking, quantum computing, and AI accelerators



AI in Industrial Automation



Automation & Efficiency

Reduces human workload.
Robotic Process Automation (RPA).



Predictive Analytics & Decision Making

AI analyzes large datasets.
Fraud detection in banking.



Personalization & Customer Experience

Based on user behavior.
Netflix Recommendation systems.



AI for Innovation & R&D

AI accelerates research.
AI-assisted medical diagnostics.

IoT in Smart Cities



Smart Infrastructure & Energy Management

Optimize resource usage.
Smart meters.



Traffic & Transportation Optimization

Reduce congestion and improve public transportation.
Smart traffic lights.



Public Safety & Environmental Monitoring

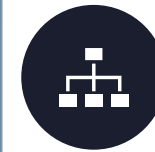
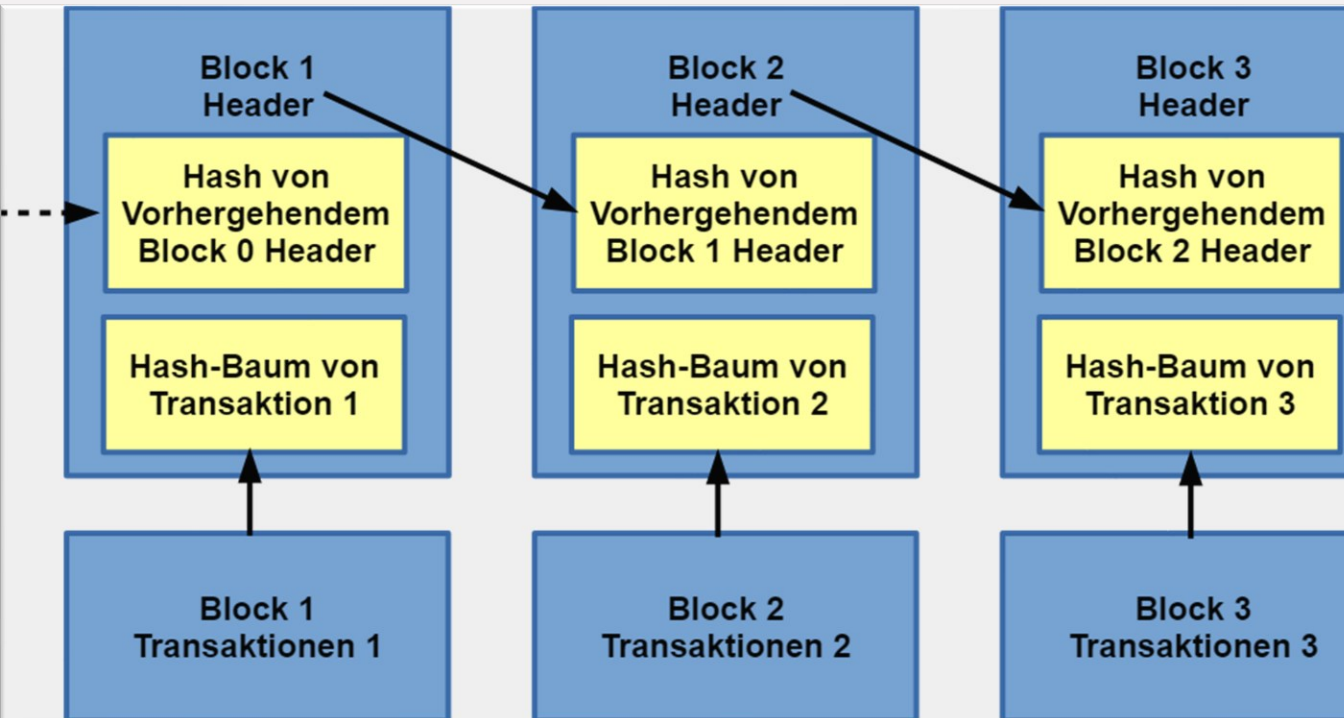
IoT sensors detect air pollution.
Connected surveillance systems.



Waste & Water Management

Optimizes waste collection routes.
Smart bins.

Block Chain for Data Security



Decentralization &
Reduced Single
Point of Failure



Immutability &
Tamper Resistance



Encryption &
Access Control



Secure Identity &
Authentication

Konvergensi Teknologi (AI, IoT, Blockchain, Cloud)



AI + IoT

- Intelligent Automation & Decision Making



Blockchain + IoT

- Secure & Transparent Data Transactions



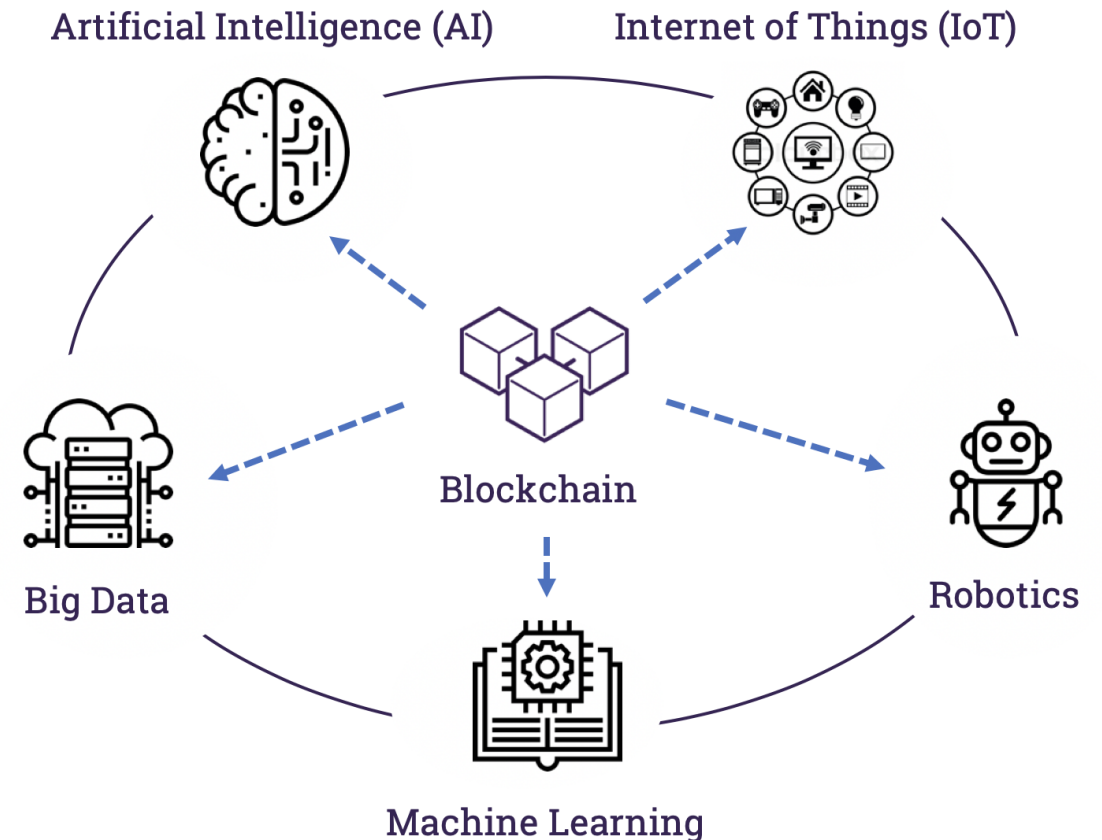
Cloud + AI

- Scalable Computing Power for AI Models



Cloud + IoT + Blockchain

- End-to-End Secure & Scalable Systems



Thank You

